

Row IV: the plus prescription, step by step

From the three-block form to evolution + DGLAP + finite, every substitution written out

$$S = k^+ + p^+, \quad \xi = \frac{k^+}{S}, \quad \bar{\xi} = 1 - \xi = \frac{p^+}{S}, \quad \xi_0 = \frac{k^+}{\sqrt{S}}, \quad r = y' - y, \quad \kappa = k_\perp \cdot r, \quad L_r = \log(4\pi^2 \mu^2 r^2), \quad \frac{1}{\epsilon} = \frac{1}{\epsilon} - \gamma_E + \log 4\pi$$

Your five-term expression is correct. It is checked below against the three-block starting point at three kinematic points; the only residual is $O(\Lambda)$, from extending the non-singular integrals from Λ down to 0, and it scales linearly in Λ . One scheme point to be aware of: doing $[\frac{1}{p^+}]_+$ in p^+ puts $\log \frac{\vee - k^+}{\Lambda}$ on the δ term, while doing it in ξ puts $\log \frac{k^+}{\Lambda}$ there. Both are right; they differ by a finite reshuffling into the $[\]_+$ remainder. §4 and §5.

1. The starting point

Abbreviate

$$\mathcal{X} = \frac{(x-y) \cdot (x'-y')}{(x-y)^2 (x'-y')^2}, \quad \mathcal{R} = \frac{(x-y) \cdot (y'-y) (x'-y') \cdot (y'-y)}{(x-y)^2 (x'-y')^2 (y'-y)^2},$$

$$\mathcal{M} = \frac{1}{\epsilon} + 1 - L_r, \quad \mathcal{U} = [U^{e'c}(y') - U^{e'c}(x')] [U^{ec}(y) - U^{ec}(x)] \rho^{e'}(x') \rho^e(x), \quad \mathcal{P} = \frac{1}{(2\pi)^3} \frac{g^4}{4\pi^3} \frac{N_c}{k^+}$$

The verified three-block form, with the phase written as $e^{-ik \cdot r} e^{-ik \cdot r p^+ / k^+} = e^{-i\kappa S / k^+}$, is

$$\frac{d^3 N^{\text{NLO}}}{d^3 k} = \mathcal{P} \int_{\Lambda}^{\vee - k^+} \frac{dp^+}{2\pi} e^{-i\kappa S / k^+} \int_{x, x', y, y'} \left\{ \underbrace{\left[\frac{p^+}{S^2} + \frac{p^+}{k^{+2}} + \frac{1}{p^+} \right] \mathcal{M} \mathcal{X}}_{(1)} - \underbrace{d_\perp \frac{p^+}{S^2} \mathcal{R}}_{(2)} - \underbrace{\left[\frac{p^+}{k^{+2}} + \frac{1}{p^+} \right] \mathcal{X}}_{(3)} \right\} \mathcal{U}$$

2. The plus prescription in p^+

Write $T = \vee - k^+$. The only singularity is $\frac{1}{p^+}$ at $p^+ \rightarrow 0$, and everything multiplying it depends on p^+ only through the phase:

$$F(p^+) = e^{-i\kappa S / k^+} = e^{-i\kappa} e^{-i\kappa p^+ / k^+}, \quad F(0) = e^{-i\kappa} \text{ finite.}$$

That is the whole condition, so

$$\int_{\Lambda}^T \frac{dp^+}{p^+} F(p^+) = \underbrace{F(0) \log \frac{T}{\Lambda}}_{\delta(p^+) \text{ piece}} + \underbrace{\int_0^T dp^+ \frac{F(p^+) - F(0)}{p^+}}_{\int_0^T dp^+ \left[\frac{1}{p^+} \right]_+ F}$$

$$\iff \frac{\theta(p^+ - \Lambda)}{p^+} = \left[\frac{1}{p^+} \right]_+ + \delta(p^+) \log \frac{\vee - k^+}{\Lambda} + O(\Lambda)$$

Applying it term by term

Block (1) contains one $\frac{1}{p^+}$, with companion $F\mathcal{M}\mathcal{X}$:

$$\mathcal{P} \int_0^T \frac{dp^+}{2\pi} F \left[\frac{p^+}{S^2} + \frac{p^+}{k^{+2}} + \left[\frac{1}{p^+} \right]_+ \right] \mathcal{M} \mathcal{X} + \mathcal{P} \frac{1}{2\pi} e^{-i\kappa} \log \frac{\vee - k^+}{\Lambda} \mathcal{M} \mathcal{X}$$

and $\mathcal{P} \cdot \frac{1}{2\pi} = \frac{1}{(2\pi)^3} \frac{g^4}{8\pi^4} \frac{N_c}{k^+}$ — your first and second blocks.

Block (2) has no pole ($\propto p^+$), so nothing happens; keep $\int_{\Lambda}$ (exact) or extend to 0 at the cost of $O(\Lambda)$ — your third block.

Block (3) contains the same $\frac{1}{p^+}$ with companion $-F\mathcal{X}$:

$$-\mathcal{P} \int_0^T \frac{dp^+}{2\pi} F \left[\frac{p^+}{k^{+2}} + \left[\frac{1}{p^+} \right]_+ \right] \mathcal{X} - \frac{1}{(2\pi)^3} \frac{g^4}{8\pi^4} \frac{N_c}{k^+} e^{-i\kappa} \log \frac{\vee - k^+}{\Lambda} \mathcal{X}$$

— your fourth and fifth blocks. Every prefactor and sign matches.

Numerical confirmation

(k^+, κ, Λ)	three-block exact	your five terms	diff
$(1.0, +0.9, 10^{-5})$	$-546.080670 - 1031.848602 i$	$-546.080786 - 1031.848694 i$	1.5×10^{-4}
$(2.5, -1.4, 2 \times 10^{-6})$	$+146.746036 + 196.977976 i$	$+146.746031 + 196.977977 i$	5.3×10^{-6}
$(0.7, +2.2, 10^{-5})$	$-60.265598 + 207.266258 i$	$-60.265879 + 207.266463 i$	3.5×10^{-4}

The residual tracks Λ linearly ($10^{-5} \rightarrow 1.5 \times 10^{-4}$, $2 \times 10^{-6} \rightarrow 5.3 \times 10^{-6}$), confirming that it is only the $\int_{\Lambda} \rightarrow \int_0$ of the regular pieces.

3. The two δ terms combine

Your second and fifth blocks have the same prefactor, the same log, the same phase $e^{-ik_{\perp} \cdot r}$ and the same $\mathcal{X}$; only the bracket differs ($\mathcal{M}$ versus 1). So

$$\frac{1}{(2\pi)^3} \frac{g^4}{8\pi^4} \frac{N_c}{k^+} \log \frac{\vee - k^+}{\Lambda} \int_{x, x', y, y'} e^{-ik_{\perp} \cdot r} [\mathcal{M} - 1] \mathcal{X} \mathcal{U}, \quad \mathcal{M} - 1 = \frac{1}{\bar{\epsilon}} - L_r$$

and that bracket is precisely $\frac{1}{\pi} \delta_{mm'} \mathcal{G}^{mm'}$, i.e. the z integral put back:

$$\frac{1}{\bar{\epsilon}} - \log(4\pi^2 \mu^2 r^2) = \frac{1}{\pi} \int d^2 z \frac{(y-z) \cdot (y'-z)}{(y-z)^2 (y'-z)^2}$$

This is the real-emission piece of the BK/JIMWLK kernel. Keeping the two blocks separate hides that.

4. Which logarithm? A scheme point, not an error

With $\bar{\xi} = \frac{p^+}{k^+ + p^+}$,

$$\log \bar{\xi} = \log p^+ - \log(k^+ + p^+) \implies \frac{d\bar{\xi}}{\bar{\xi}} = \frac{dp^+}{p^+} - \frac{dp^+}{k^+ + p^+} = \xi \frac{dp^+}{p^+}$$

so $[\frac{1}{p^+}]_+$ and $[\frac{1}{1-\xi}]_+$ are *not* the same distribution. Their δ coefficients are

$$\text{in } p^+ : \log \frac{\vee - k^+}{\Lambda}, \quad \text{in } \xi : \log \frac{1 - \xi_0}{1 - \xi_1} \xrightarrow{\Lambda \ll k^+ \ll \vee} \log \frac{k^+}{\Lambda}$$

differing by $\log \frac{\vee - k^+}{k^+}$, which is reshuffled into the $[\]_+$ remainder. The total is identical (both were checked against the same three-block integral). Pick whichever matches the rapidity variable in your evolution equation; for matching onto DGLAP the ξ version is the natural one, and the rest of this note uses it.

5. Longitudinal momentum fraction

The $\mathcal{C}_1$ vertex splits a parent of $S = k^+ + p^+$ into the measured gluon (k^+) and an unmeasured one (p^+). The measured fraction is $\xi = k^+/S$, so

$$p^+ = k^+ \frac{1 - \xi}{\xi}, \quad dp^+ = -\frac{k^+}{\xi^2} d\xi, \quad \frac{S}{k^+} = \frac{1}{\xi}, \quad \begin{cases} p^+ = \Lambda & \rightarrow \xi_1 = \frac{k^+}{k^+ + \Lambda} \\ p^+ = \vee - k^+ & \rightarrow \xi_0 = \frac{k^+}{\vee} \end{cases}$$

Three consequences, each a fragmentation signature:

$$\begin{aligned} \text{(i)} \quad & \frac{p^+}{S^2} = \frac{\xi \bar{\xi}}{k^+}, \quad \frac{p^+}{k^{+2}} = \frac{1}{k^+} \frac{\bar{\xi}}{\xi}, \quad \frac{1}{p^+} = \frac{1}{k^+} \frac{\xi}{\bar{\xi}} \\ \implies & \frac{p^+}{S^2} + \frac{p^+}{k^{+2}} + \frac{1}{p^+} = \frac{1}{k^+} \left[\xi \bar{\xi} + \frac{\bar{\xi}}{\xi} + \frac{\xi}{\bar{\xi}} \right] = \frac{P_{gg}(\xi)}{2N_c k^+} \end{aligned}$$

— your bracket is the gluon splitting function, term by term.

$$\text{(ii)} \quad \int_{\Lambda}^{\vee - k^+} dp^+ = k^+ \int_{\xi_0}^{\xi_1} \frac{d\xi}{\xi^2} \quad \text{— the fragmentation measure } \frac{d\xi}{\xi^2}$$

$$(iii) \quad e^{-ik \cdot r} e^{-ik \cdot r p^+ / k^+} = e^{-i \frac{k_\perp}{\xi} \cdot r} \quad \text{--- the hard factor at } k_\perp / \xi$$

The Wilson lines sit at x, x', y, y' with no ξ dependence, which is the fourth signature. The Jacobian's k^+ cancels the $\frac{1}{k^+}$ of (i), so the prefactor $\mathcal{P}$ is unchanged. Written ascending in ξ (so the signs are the same as in p^+):

$$\frac{d^3 N^{\text{NLO}}}{d^3 k} = \mathcal{P} \int_{\xi_0}^{\xi_1} \frac{d\xi}{2\pi} \frac{1}{\xi^2} \int_{x, x', y, y'} e^{-i\kappa/\xi} \left\{ \mathcal{C}(\xi) \mathcal{M} \mathcal{X} - d_\perp \xi \bar{\xi} \mathcal{R} - \left[\frac{\xi}{\xi} + \frac{\bar{\xi}}{\xi} \right] \mathcal{X} \right\} \mathcal{U}, \quad \mathcal{C} = \frac{P_{gg}}{2N_c}$$

6. The plus prescription in ξ

Only $\frac{\xi}{\xi}$ is singular at $\xi \rightarrow 1$, and it appears twice: inside $\mathcal{C}$ times $\mathcal{M}$, and in the last block times -1 . They combine:

$$\frac{1}{\xi^2} e^{-i\kappa/\xi} [\mathcal{M} - 1] \frac{\xi}{1 - \xi} \mathcal{X} = \frac{h(\xi)}{1 - \xi} [\mathcal{M} - 1] \mathcal{X}, \quad h(\xi) \equiv \frac{e^{-i\kappa/\xi}}{\xi}, \quad h(1) = e^{-i\kappa}$$

$h(1)$ is finite, so

$$\int_{\xi_0}^{\xi_1} d\xi \frac{h(\xi)}{1 - \xi} = h(1) \log \frac{1 - \xi_0}{1 - \xi_1} + \int_{\xi_0}^{\xi_1} d\xi \frac{h(\xi) - h(1)}{1 - \xi} \longrightarrow e^{-i\kappa} \log \frac{k^+}{\Lambda} + \int_{\xi_0}^1 d\xi \frac{h(\xi) - h(1)}{1 - \xi}$$

Define the regularised splitting function by exactly that subtraction:

$$\hat{\mathcal{C}}(\xi) = \xi \bar{\xi} + \frac{\bar{\xi}}{\xi} + \frac{\xi}{[1 - \xi]_+} = \frac{\hat{P}_{gg}(\xi)}{2N_c}, \quad \int_{\xi_0}^1 \frac{d\xi}{\xi^2} e^{-i\kappa/\xi} \frac{\xi}{[1 - \xi]_+} \equiv \int_{\xi_0}^1 d\xi \frac{h(\xi) - h(1)}{1 - \xi}$$

Everything else — $\xi \bar{\xi}$, $\frac{\bar{\xi}}{\xi}$, $\mathcal{M}$, $\mathcal{X}$, $\mathcal{R}$ and the Wilson lines — is regular at $\xi = 1$ and is untouched.

7. Regrouping into three sectors

Split $\mathcal{M} = \frac{1}{\bar{\epsilon}} + 1 - L_r$ and $\mathcal{M} - 1 = \frac{1}{\bar{\epsilon}} - L_r$, then collect:

- the $\delta(1 - \xi)$ term (from §6) carries $[\frac{1}{\bar{\epsilon}} - L_r] \log \frac{k^+}{\Lambda}$;
- the $\frac{1}{\bar{\epsilon}}$ from $\mathcal{M}$ multiplies $\xi \bar{\xi} + \frac{\bar{\xi}}{\xi}$, and the $\frac{1}{\bar{\epsilon}}$ from $\mathcal{M} - 1$ multiplies the $[\]_+$ remainder — together exactly $\frac{1}{\bar{\epsilon}} \hat{\mathcal{C}}(\xi)$;
- the remaining $1 - L_r$ pieces plus the $-\frac{\bar{\xi}}{\xi} \mathcal{X}$ and $-d_\perp \xi \bar{\xi} \mathcal{R}$ collapse, using $[\xi \bar{\xi} + \frac{\bar{\xi}}{\xi}](1 - L_r) - \frac{\bar{\xi}}{\xi} = \xi \bar{\xi} - L_r [\xi \bar{\xi} + \frac{\bar{\xi}}{\xi}]$, into $-L_r \hat{\mathcal{C}}(\xi) \mathcal{X} + \xi \bar{\xi} [\mathcal{X} - d_\perp \mathcal{R}]$.

Since $\frac{1}{\bar{\epsilon}} \rightarrow 1$ and $e^{-i\kappa/\xi} \rightarrow e^{-i\kappa}$ at $\xi = 1$, all three fit under one integral.

$$\frac{d^3 N^{\text{NLO}}}{d^3 k} = \frac{1}{(2\pi)^3} \frac{g^4}{8\pi^4} \frac{N_c}{k^+} \int_{\xi_0}^1 \frac{d\xi}{\xi^2} \int_{x, x', y, y'} e^{-i \frac{k_\perp}{\xi} \cdot (y' - y)} \left\{ \mathcal{E} + \mathcal{D} + \mathcal{F} \right\} \mathcal{U}$$

$$\mathcal{E} = \delta(1 - \xi) \log \frac{k^+}{\Lambda} \left[\frac{1}{\bar{\epsilon}} - L_r \right] \mathcal{X} \quad \textbf{evolution}$$

$$\mathcal{D} = \frac{1}{\bar{\epsilon}} \hat{\mathcal{C}}(\xi) \mathcal{X} \quad \textbf{DGLAP}$$

$$\mathcal{F} = -L_r \hat{\mathcal{C}}(\xi) \mathcal{X} + \xi(1 - \xi) [\mathcal{X} - d_\perp \mathcal{R}] \quad \textbf{finite}$$

Verification

Sum of the three sectors against the exact ξ integrand, upper limit kept at ξ_1 so nothing is dropped:

(k^+, κ)	exact	$\mathcal{E} + \mathcal{D} + \mathcal{F}$	diff
(1.0, +0.9)	826.912590 + 338.307363 i	826.912590 + 338.307363 i	2.7×10^{-14}
(2.5, -1.4)	-150.161862 + 96.165817 i	-150.161862 + 96.165817 i	4.5×10^{-15}
(0.7, +2.2)	-36.595454 + 176.960721 i	-36.595454 + 176.960721 i	2.8×10^{-14}

8. What each sector is

Evolution

$$\left. \frac{d^3 N}{d^3 k} \right|_{\text{evol}} = \frac{1}{(2\pi)^3} \frac{g^4}{8\pi^5} \frac{N_c}{k^+} \log \frac{k^+}{\Lambda} \int_{x,x',y,y'} e^{-ik_\perp \cdot (y'-y)} \frac{(x-y) \cdot (x'-y')}{(x-y)^2 (x'-y')^2} \int d^2 z \frac{(y-z) \cdot (y'-z)}{(y-z)^2 (y'-z)^2} \mathcal{U}$$

The hard scale is $k_\perp$, not $k_\perp/\xi$ — at $\xi = 1$ nothing is emitted into the measured direction. The z integral is the real-emission BK/JIMWLK kernel, and $\log \frac{k^+}{\Lambda}$ is the rapidity interval. **It still carries $\frac{1}{\epsilon}$** , which is that kernel's own large- $|z|$ divergence: $\int^\infty \frac{d^2 z}{z^2}$. It cancels only when the Group II rows in which $U(z)$ survives the colour reduction are added, completing $\frac{(y-y')^2}{(y-z)^2 (y'-z)^2}$, which is large- z convergent. Do not absorb it into a fragmentation function — an FF cannot depend on Λ , and this term does.

DGLAP

$$\begin{aligned} \left. \frac{d^3 N}{d^3 k} \right|_{\text{DGLAP}} &= \frac{1}{(2\pi)^3} \frac{g^4}{8\pi^4} \frac{1}{k^+} \frac{1}{\bar{\epsilon}} \int_{\xi_0}^1 \frac{d\xi}{\xi^2} \frac{\widehat{P}_{gg}(\xi)}{2} \int_{x,x',y,y'} e^{-i\frac{k_\perp}{\xi} \cdot (y'-y)} \frac{(x-y) \cdot (x'-y')}{(x-y)^2 (x'-y')^2} \mathcal{U} \\ &= \frac{\alpha_s}{2\pi} \frac{1}{\bar{\epsilon}} \int \frac{d\xi}{\xi^2} \frac{\widehat{P}_{gg}(\xi)}{2} \frac{d^3 N^{\text{LO}}}{d^3 k} \left(\frac{k_\perp}{\xi} \right), \quad \widehat{P}_{gg}(\xi) = 2N_c \left[\xi \bar{\xi} + \frac{\bar{\xi}}{\xi} + \frac{\xi}{[1-\xi]_+} \right] \end{aligned}$$

(the $\frac{\alpha_s}{2\pi}$ follows from $\frac{g^4/8\pi^4}{g^2/\pi^2} = \frac{g^2}{8\pi^2}$; the residual $\frac{1}{2}$ on $\widehat{P}_{gg}$ depends on the LO normalisation convention and on whether the hermitian-conjugate row is counted separately — fix it against your own LO.) Remove the pole with

$$D_{g/g}(\xi, \mu_F) = \delta(1-\xi) + \frac{\alpha_s}{2\pi} \left[-\frac{1}{\bar{\epsilon}} + \log \frac{\mu_F^2}{\mu^2} \right] \widehat{P}_{gg}(\xi) + O(\alpha_s^2)$$

which sends $\mu \rightarrow \mu_F$ in the finite sector.

Finite

$$\left. \frac{d^3 N}{d^3 k} \right|_{\text{fin}} = \frac{1}{(2\pi)^3} \frac{g^4}{8\pi^4} \frac{N_c}{k^+} \int_{\xi_0}^1 \frac{d\xi}{\xi^2} \int_{x,x',y,y'} e^{-i\frac{k_\perp}{\xi} \cdot (y'-y)} \left\{ -\log(4\pi^2 \mu_F^2 (y'-y)^2) \widehat{\mathcal{C}}(\xi) \mathcal{X} + \xi \bar{\xi} [\mathcal{X} - d_\perp \mathcal{R}] \right\} \mathcal{U}$$

Two structures only. The first is $\widehat{P}_{gg}$ riding on the position-space logarithm: this is the NLO hard factor, and it does *not* factor out of the LO because $r = y' - y$ is integrated. The second is the pure transverse tensor

$$\xi \bar{\xi} X^m X'^m \left[\delta^{mm'} - d_\perp \hat{r}^m \hat{r}^{m'} \right], \quad X = \frac{x-y}{(x-y)^2}, \quad X' = \frac{x'-y'}{(x'-y')^2}, \quad \hat{r} = \frac{y'-y}{|y'-y|}$$

the azimuthal partner of the $\xi \bar{\xi}$ term of P_{gg} : no pole, no $\frac{1}{1-\xi}$, no logarithm, and it vanishes on averaging over $\hat{r}$ at $d_\perp = 2$.

9. Loose ends

- **Where Λ comes from.** $\Theta(k_{\text{arg}}^+ - p_{\text{arg}}^+ - \Lambda)$ in your $\mathfrak{C}_{1jki}^{bca}$ becomes $\Theta(k^+ - \Lambda)$ in this row — a condition on the measured gluon, always satisfied. The $\frac{1}{p^+}$ pole needs its own $\Theta(p^+ - \Lambda)$; settle this before quoting the $\log \frac{k^+}{\Lambda}$ coefficient against other rows.
- **The small- ξ end.** $\widehat{P}_{gg} \rightarrow \frac{2N_c}{\xi}$ against $\frac{d\xi}{\xi^2}$ is a ξ^{-3} integrand at $\xi \rightarrow \xi_0 = \frac{k^+}{\sqrt{V}}$, held only by the oscillation of $e^{-ik_\perp \cdot r/\xi}$. Do the y, y' integrals before sending $V \rightarrow \infty$.
- **Still to fix upstream:** in (0.2) the overall $\frac{k^+ + p^+}{k^+}$ is spurious (prefactor is $\frac{1}{k^+}$) and that factor belongs in the exponent, as (0.3) has it; in (0.18)/(0.19) $\mu^{2\epsilon} \rightarrow \mu^{-2\epsilon}$; delete $\int_{w,w'}; \int_{x,y,y'} \rightarrow \int_{x,x',y,y'}; U^{ce}(x) \rightarrow U^{ec}(x)$.

Checks: `src/rowG2_pplus_split.py` (the five-term p^+ form), `src/rowG2_xicheck.py` (the change of variable), `src/rowG2_dglap.py` (the P_{gg} identity and the $\frac{\alpha_s}{2\pi}$ normalisation), `src/rowG2_sectors.py` (the three-sector sum), `src/rowG2_vertex.py` (eq. (0.1) from the vertex definitions).